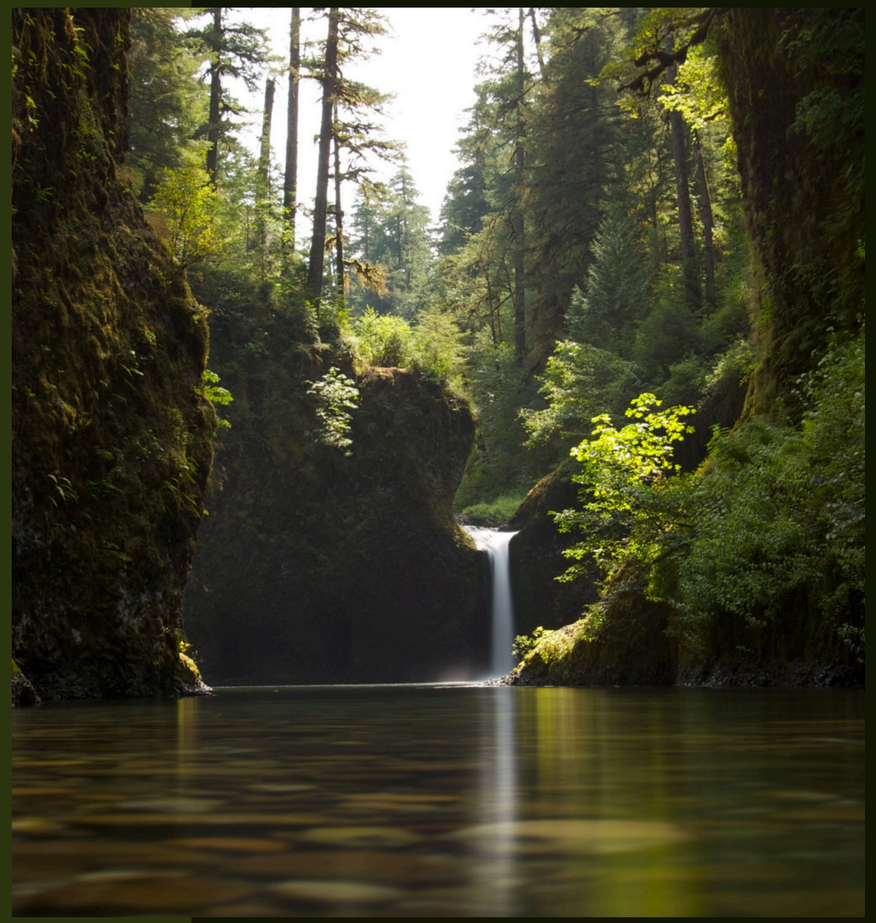
IE0005 Final Project

Detecting Cardiovascular Disease
with Non-Invasive Features

Can machine learning beat the clinical blood-pressure rule, and does it work equally well across age and sex?

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Cardiovascular Disease (Kaggle, 70,000 patients)



Can non-invasive features beat the clinical blood-pressure rule?

CVD screening often relies on a simple threshold: systolic ≥ 140 or diastolic ≥ 90 mmHg. Lab tests such as cholesterol and glucose are invasive and costly.

Question 1 **Detection**

Do age, sex, height, weight, blood pressure and lifestyle detect CVD better than the clinical rule?

Question 2 **Fairness**

Is detection equally effective across age and sex groups? A screening tool should not fail one group of patients.

Problem type: binary classification (cardio = 0 / 1)

70,000 patients and a balanced target

70,000
patients

11
features + 1 target (cardio)

0
missing values

50/50
no CVD / CVD

Feature group	Features
Objective	age, gender, height, weight
Examination	systolic BP, diastolic BP, cholesterol, glucose
Subjective	smoking, alcohol intake, physical activity



Source: Kaggle, Cardiovascular Disease dataset

Data errors to fix, and a strong age effect

Impossible blood pressure

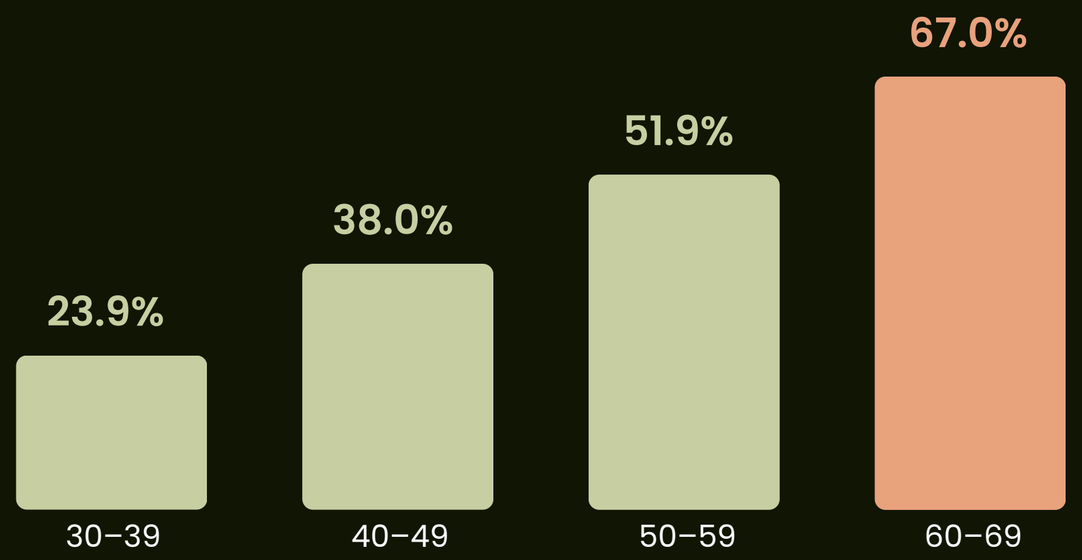
Max systolic 16,020 mmHg, negative values, and 1,236 rows with diastolic ≥ systolic.

Sex makes little difference

Prevalence 49.7% (women) vs 50.5% (men).

Cholesterol matters

Prevalence 44.0% → 60.2% → 76.5% across levels 1–3.



Cleaning, new features and a stratified split

1 · Remove implausible values

Kept systolic 70–240, diastolic 40–150, systolic > diastolic, height 120–220 cm, weight 30–200 kg. 1,390 rows removed (2%), 68,610 kept.

2 · Create new features

BMI = weight / height², pulse pressure = systolic - diastolic, and 10-year age groups.

3 · Keep features non-invasive

Cholesterol and glucose are excluded, so the model only uses measurements taken without a lab test.

4 · Split without resampling

80% train / 20% test, stratified by the target. Classes are already 50/50, so no balancing is needed.

Three methods, one held-out test set

Baseline

Clinical rule

Predicts CVD if systolic ≥ 140 or diastolic ≥ 90. No training needed.

From the course

Decision Tree

max_depth = 5. Easy to read: each split is a yes/no question.

Beyond the course

Random Forest

100 trees, max_depth = 8, majority vote. More stable than one tree.

How we evaluate

All three methods are tested on the same 20% held-out test set (13,722 patients) using accuracy, precision, recall and F1-score. Recall is then compared across age and sex groups to check whether detection is balanced.

What we learned outside the syllabus

Random Forest

100 decision trees, each trained on random rows and features. The final answer is a majority vote.

Precision, recall and F1-score

Evaluation metrics beyond the accuracy and confusion matrix covered in the course.

Clinical baseline

Benchmarking ML against a real screening rule, not only against another model.

Subgroup recall analysis

Checking whether detection stays balanced across age and sex groups.

Both ML models detect more CVD cases than the clinical rule

Method	Accuracy	Precision	Recall	F1-score
Clinical rule	0.699	0.777	0.548	0.643
Decision Tree	0.717	0.732	0.676	0.703
Random Forest	0.722	0.748	0.659	0.701

Higher recall

ML detects 11–13 points more CVD cases than the clinical rule (Decision Tree 67.6% vs 54.8%).

Best accuracy and F1

Random Forest has the highest accuracy (72.2%); Decision Tree has the highest F1-score (0.703).

Lower precision

The clinical rule is the most precise (0.777): ML flags more patients who turn out to be healthy.

Recall is balanced across sexes, but rises with age

Recall by age group (test set)

Group	n	Clinical rule	Decision Tree	Random Forest
30–39	404	0.581	0.595	0.568
40–49	3,788	0.529	0.56	0.566
50–59	7,049	0.551	0.652	0.63
60–69	2,480	0.556	0.831	0.807

Recall by sex (test set)

Group	n	Clinical rule	Decision Tree	Random Forest
Women	9,027	0.532	0.671	0.653
Men	4,695	0.579	0.685	0.669

ML’s recall advantage grows with age

Under 50

56–60%

ML recall vs 53–58% for the clinical rule

Close to the clinical rule

Ages 50-59

63–65%

max_depth = 5. Easy to read: each split is a yes/no question.

+8 to +10 points

Ages 60-69

81–83%

100 trees, max_depth = 8, majority vote. More stable than one tree.

+25 to +28 points

Non-invasive ML is most useful for screening older adults. Under 50, lab markers such as cholesterol and glucose may be needed.

Non-invasive ML beats the clinical rule, especially for older patients

Objective 1 · Met

Both ML models have higher accuracy, recall and F1-score than the clinical rule, with 11–13 points more recall.

Objective 2 · Partly met

Recall is balanced across sexes, but the ML advantage is concentrated in patients aged 50 and over.

Trade-off

ML flags more healthy patients than the clinical rule (precision 0.73–0.75 vs 0.78).

Next steps

Test adding cholesterol and glucose for younger patients, and use cross-validation instead of a single 80/20 split.

Thankyou!

Member	Contribution
Michelle Aline Lorenzo	problem & objective, dataset
Jovica Marcia Sugianto	overall results, subgroup recall
Soh Ting Hui Chloe	key finding, conclusion
Tan Boon Han	ML approach, beyond the course
Vuu Tan Loc	EDA, data preparation